



**Hochschule
Zittau/Görlitz**
UNIVERSITY OF APPLIED SCIENCES

Master's Thesis

Something with XAI stuff and making it actually work

Master's Program in Computer Science

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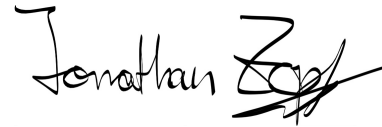
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Görlitz, June 29, 2026

Declaration of Originality

I hereby declare that I have written this thesis independently and have not used any sources other than those indicated. All passages that are taken verbatim or in spirit from published or unpublished sources are clearly marked as such. This work has not been submitted to any other examination board in the same or similar form.

Görlitz, June 29, 2026

A handwritten signature in black ink, appearing to read "Jonathan Zepf". The signature is stylized with a large, sweeping 'Z' and a final flourish.

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Chapter 1

Introduction

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Chapter 2

Background

2.1 Introduction to LLM Explainability

2.1.1 Definition and Purpose

Large language models (LLMs) have seen increasing adoption in recent years for tasks such as question answering [3], writing [18], and software development [42], but also in high-stakes domains such as healthcare [23, 16, 13]. However, since LLMs are complex deep learning models with millions of neurons and billions of parameters, they are often considered black boxes [30]. This makes it challenging to comprehend and evaluate their decisions.

Explainable AI (XAI) is a research field that seeks to make artificial intelligence algorithms transparent and understandable [12], thereby making the inherent black box less opaque. Central to this research field is the concept of explainability, whose precise definition remains disputed [33]. The term is described in the relevant ISO standard as the “level of understanding of how the AI-based system came up with a given result” [14]. Explainability aims to justify decisions and demonstrate that they are correct and unbiased in order to build trust with users [2].

The term *interpretability* is often used in a similar context, although there is a subtle distinction between the two concepts. Interpretability concerns understanding the inner workings of an AI model, whereas explainability focuses on making the decision process retraceable [8, 2].

2.1.2 Transparency as Regulatory Requirement

The recent increase in the importance of XAI research has also been driven by new regulatory measures. While multiple countries have introduced their own legislative frameworks to govern the use of AI, such as China [45] and the United States [5], the European Union’s AI Act [31] can be regarded as particularly relevant to explainability. Under this regulation, individuals affected by high-risk AI decisions have the right to obtain clear and meaningful explanations regarding the AI system’s role and the main contributing factors to the decision. Furthermore, high-risk AI systems must provide transparency and allow for human oversight [31]. The *Right to Explanation*

derived from the General Data Protection Regulation (GDPR) is also relevant to AI systems within the European Union [27].

2.1.3 Audience Groups

Providing explanations for reasoning processes or decisions made by an AI system can increase trust in the system among end users and experts [40, 9], while also assisting researchers and developers in debugging and fine-tuning the model [37]. Moreover, as argued in Chapter 2.1.2, explainability can be required by regulations and auditors, thereby forming another stakeholder group for XAI systems [9, 37].

It has also been noted that different audiences perceive and consume explanations differently [10, 9]. For instance, developers may expect technical information about the inner workings of the system, whereas end users may benefit more from concise and understandable explanations in natural language. Adapting explanations generated by an AI system to the expectations of the recipient has therefore become a subject of recent research [11, 25].

2.1.4 Aspects and Dimensions

Herrera [9] points out that, whereas traditional AI models usually follow fixed deterministic rules when generating explanations, LLMs complicate this process due to their more complex and advanced capabilities. The author therefore proposes four dimensions of explainability for large language models:

- **Faithfulness:** Faithfulness describes whether the explanation corresponds to the internal reasoning of the model [15].
- **Truthfulness:** Truthfulness describes whether the explanation conforms to real-world facts derived from an external authoritative reference. A truthful explanation does not contain factual errors, omissions, or distortions [9].
- **Plausibility:** Plausibility describes the degree to which an explanation is convincing and appears reasonable to humans [44, 15].
- **Contrastivity:** Contrastivity measures how well the explanation highlights differences between competing pieces of information. In classification tasks, an XAI system with high contrastivity clearly outlines *why* it selected one class over another [9, 26, 35].

It should be noted that there is still no broad scientific consensus within the XAI community regarding which aspects are essential for an explanation to be considered good, since descriptions, terminology, and definitions of these factors vary considerably [29, 2, 19, 26]. This issue is further amplified by regulatory and standardization frameworks, which remain comparatively vague in this regard [31, 14]. Nevertheless, Herrera’s [9] desiderata of faithfulness and plausibility in particular are supported by other literature, including [15, 44, 38]. Other authors, such as Lopes et al. [19], refer to faithfulness as *fidelity* and divide plausibility into several human-centered sub-aspects.

2.1.5 LLM Explainability Challenges

According to Zhao et al. [43, 21], explainability research for LLMs is difficult because understanding their internal processes becomes infeasible due to their enormous size and complex structure. Traditional methods such as gradient-based attribution [36] and SHAP values [20] are impractical for state-of-the-art LLMs with billions of parameters. This complexity also hinders in-depth analysis, debugging, and diagnosis of these models. Furthermore, the opaque black-box nature of LLMs amplifies this problem [30]. Many mainstream commercial models are only accessible via APIs [24], meaning that their model weights are not publicly available to users or researchers.

Despite these limitations, several technical approaches have been developed to gain insight into the black box and retrace important reasoning pathways within an LLM. These approaches cannot be discussed in detail within the scope of this thesis, but overviews of the most prominent methods are provided in [43, 21].

At the same time, since LLMs are specifically designed for natural language processing tasks, they also provide an opportunity to generate explanations directly in natural human language [43]. Most notably, this includes *ad-hoc* explanations where the model generates a justification for its decision after giving the prediction and *Chain-of-Thought* (CoT) explanations, in which LLMs reconstruct their own reasoning process. Prompting an LLM to generate CoT explanations has furthermore been shown to improve performance on complex reasoning tasks [41].

2.2 Faithfulness of LLM Explanations

2.2.1 Definition and Significance

Faithfulness describes the degree to which the explanation generated by an AI-powered system is grounded in the model’s actual internal reasoning process [9, 15, 32]. A faithful explanation should therefore reflect the model’s true decision path rather than merely providing a plausible post-hoc rationalization [9].

LLMs have the ability to give those explanations in natural language [41]. But unlike other explainability methods, the faithfulness of these are not guaranteed [9, 15]. In particular, an LLM may generate a natural language explanation that sounds convincing to users (high plausibility) while being entirely fabricated and not grounded in the model’s actual reasoning process (low faithfulness). Such scenarios can pose significant risks and lead to misplaced trust in these systems [1, 22, 9].

In particular, research by Turpin et al. [38] demonstrated that LLMs can systematically conceal internalized biases while providing plausible yet unfaithful justifications. When deliberately influenced to rely on prejudicial reasoning in their experiments, the models often used social biases such as gender or race as primary decision factors, but rarely mentioned them explicitly in their explanations. Instead, they resorted to alternative justifications based on conduct or circumstances. This behavior is especially concerning in applications such as recidivism prediction, where fairness may be undermined [7, 15]. Consequently, it is crucial not only to verify

whether an explanation appears reasonable, but also whether it accurately reflects the actual reasoning process of the model, particularly in high-stakes domains.

2.2.2 Estimation Approaches

General Challenges

As discussed in Section 2.1.5, the complex and opaque nature of large language models poses unique challenges for explainability research. These challenges are particularly pronounced when assessing the faithfulness of CoT explanations. Whereas plausibility can be evaluated by human judges [9], faithfulness is inherently tied to the model’s internal reasoning process [28, 15].

Faithfulness vs Self-Consistency

Some have argued such as [28] that faithfulness can not be measured and only self-consistency can be. Present here the case, why we still talk about measuring or estimating faithfulness.

Early Faithfulness Work

One of the earliest works addressing the estimation of faithfulness in LLMs was conducted by Turpin et al. [38]. In addition to demonstrating that CoT explanations can systematically omit contributing decision factors, the authors proposed a quantification approach for unfaithfulness on the BBH dataset. Their method measures the drop in model accuracy when biasing features are added to inputs that the model fails to mention in its explanations. However, the approach is labor-intensive, as CoT explanations generated by the LLM must be manually inspected by human annotators to determine whether the biased feature is mentioned. According to the authors, the method only evaluates faithfulness with respect to relatively small input modifications, whereas a practically useful metric should also be able to investigate reasoning behavior over larger input changes.

Another early contribution was presented by Chen et al. [6], who introduced the concept of *counterfactual simulatability*. This concept describes the degree to which humans can infer “the model’s outputs on diverse counterfactuals of the explained input.” In other words, after reading an explanation to an initial question, a human should be able to predict how the model will answer closely related counterfactual yes-or-no questions. Based on this concept, the authors propose two metrics: *simulation generality*, which measures the diversity of counterfactuals relevant to the explanation, and *simulation precision*, which measures the extent to which human expectations align with the model’s actual outputs on those counterfactuals.

Their evaluation pipeline first selects a question from a dataset and prompts the LLM to generate both an answer and an explanation. This explanation is subsequently used to generate corresponding counterfactual questions, a task performed by another LLM in order to scale to larger datasets. Human evaluators then construct a mental model based on the original explanation and state their expectations

regarding the answers to the counterfactual questions. Finally, the counterfactuals are presented to the LLM and the generality and precision scores are computed. This approach is notable because it supports both ad-hoc explanations and CoT explanations. It also represents an early example of using auxiliary LLMs to generate counterfactuals, thereby inheriting the limitations associated with LLM-as-a-Judge approaches [24].

Counterfactual Edits

Another influential use of AI-generated counterfactuals was introduced by Atanasova et al. [4]. The authors laid the foundation for a line of work commonly referred to as *counterfactual edits* [28, 24], which has subsequently been adopted and extended by later metrics [34, 24]. In these approaches, the LLM’s response to a counterfactual input serves as the basis for estimating faithfulness.

For their method, Atanasova et al. trained an auxiliary model to insert words into the original input, thereby generating counterfactuals. Both the original input and the modified version are then provided to the evaluated LLM. The resulting predictions and the explanation for the counterfactual input are subsequently analyzed. If the prediction changes due to the inserted words and those words are not mentioned in the explanation, the explanation is considered unfaithful. A major advantage of this approach is that it does not require human annotators [28].

Nevertheless, several limitations remain. The inserted modifications may shift the model’s attention to unrelated aspects of the input, meaning that the prediction change is not necessarily caused by the inserted feature itself [4]. Siegel et al. [34] additionally identify two practical shortcomings of this implementation of counterfactual edits. First, the method does not investigate whether highly influential features are more likely to be mentioned than weakly influential ones. In the extreme case, an AI system could theoretically achieve perfect faithfulness simply by repeating the entire input text. Second, the binary impact measurement only evaluates whether the prediction probability crosses the 50% threshold, thereby ignoring potentially meaningful probability shifts that remain below this boundary.

Correlation-Based Metrics

To address these limitations, Siegel et al. [34] proposed the Correlational Counterfactual Test (CCT), which significantly extends the approach of Atanasova et al. [4]. At its core, CCT measures faithfulness as the correlation between intervention impact scores and explanation mention scores [24].

First, the LLM receives an unmodified question from a dataset. The input is then perturbed and presented to the LLM again, producing prediction probability distributions for both the original and counterfactual versions. The counterfactual generation procedure largely follows the earlier work by Atanasova et al., although an auxiliary LLM is additionally used to filter nonsensical interventions. The Total Variation Distance (TVD) between the original and perturbed prediction distributions is then calculated in order to quantify the impact of the intervention. Subsequently, the

generated explanation is analyzed using substring matching to determine whether the intervention is explicitly mentioned, resulting in a binary explanation score. This procedure is repeated across multiple dataset examples, after which the Pearson correlation coefficient between intervention impacts and explanation mentions is computed to obtain the final CCT score.

Since the metric builds directly upon the approach by Atanasova et al. [4], several limitations are inherited, particularly with respect to counterfactual generation. For example, the counterfactuals are restricted to the insertion of individual adjectives or adverbs, limiting their semantic diversity. Furthermore, the metric does not account for cases in which an explanation refers to a synonym or semantically equivalent concept rather than the exact inserted word.

Concept-Based Metrics

Building upon the idea of counterfactual edits, Matton et al. [24] proposed the *Causal Concept Faithfulness Metric*. Unlike previous approaches, their method operates on higher-level concepts such as *gender*, *race* or *time of day* instead of low-level tokens or words. Furthermore, the method not only produces a quantitative score but also exposes semantic patterns related to faithfulness.

The authors define faithfulness as the alignment between truly influential concepts in the model’s prediction and the importance assigned to those concepts by the model itself within its explanation. To estimate the causal influence of concepts, an auxiliary LLM either replaces or removes concept values from the original input. As in the CCT metric, the original and counterfactual questions are then presented to the LLM and prediction distributions are collected, yielding estimates of the true causal effect. Subsequently, another helper LLM extracts the concepts deemed important within the explanation, enabling the calculation of the explanation-implied effect. The Pearson correlation between the true causal effects and the explanation-implied effects then serves as the final faithfulness score.

One limitation of this approach lies in the LLM-based automation of counterfactual generation and explanation extraction, which may introduce errors and risks of circularity within the evaluation process. Furthermore, the metric does not perform well when concepts are strongly correlated. Since this metric is central to the present thesis, it will be discussed in greater technical detail in Chapter 2.3.

Output Contribution Metrics

Parcalabescu and Frank [28] argue that many existing metrics measure self-consistency of outputs rather than genuine faithfulness. They therefore propose their own self-consistency metric, CC-SHAP, which does not depend on modifying the input. Their approach compares the degree to which input tokens contribute to both the generated answer and the explanation. By measuring the convergence between these contributions using Shapley values, the method produces a continuous score.

This approach offers several advantages. It avoids the need for input perturbations, does not require semantic evaluation of explanations, supports both CoT and

ad-hoc explanations, reveals patterns of consistency and inconsistency at the token level, and does not require auxiliary AI models during evaluation. However, these benefits come at the cost of considerable computational overhead, which may exceed that of other approaches [28].

CoT Corruption Methods

Lanham et al. [17] investigated faithfulness by deliberately corrupting CoT reasoning in order to test hypotheses regarding CoT behavior. First, the model is prompted to reason step-by-step about a multiple-choice problem. Given the original CoT, the LLM then produces a final answer. The CoT is subsequently modified through several interventions, including truncation, the insertion of mistakes using an auxiliary model, paraphrasing of earlier reasoning steps followed by regeneration of the remainder, and the insertion of filler tokens. The model is then asked again to provide an answer based on the modified CoT.

If the prediction remains unchanged despite the intervention, the explanation is considered unfaithful. Since the metric primarily serves as a tool for evaluating research hypotheses about CoT reasoning, it remains unclear how directly comparable it is to other faithfulness metrics. Parcalabescu and Frank [28] further argue that this approach assumes that the model fundamentally relies on the CoT in order to produce the correct prediction. However, experiments conducted by Lanham et al. [17] themselves suggest that CoT explanations only marginally improve prediction accuracy compared to models without CoT reasoning [28], thereby raising questions regarding the practical usefulness of this method.

Faithfulness Through Unlearning

A final approach is presented by Tutek et al. [39], who propose a novel method called *Faithfulness by Unlearning Reasoning Steps* (FUR). The central idea is that if a reasoning step is genuinely used by the model to derive its answer, then removing the corresponding information from the model’s parameters should alter its prediction.

For a given LLM M and a test question, the model first generates a CoT explanation. The explanation is then decomposed into individual reasoning steps. For each step, the corresponding knowledge is selectively unlearned from the model parameters, producing a fine-tuned model M' that is discouraged from predicting the removed reasoning step in the given context. Subsequently, both the original model M and the modified model M' are asked the same question again, this time without requesting a CoT explanation, and their predictions are compared.

For the proposed ff-hard metric, which evaluates the faithfulness of the entire CoT, the result is binary. A score of 1 indicates that the models disagree on the most likely answer, suggesting high faithfulness, whereas a score of 0 indicates that the prediction remains unchanged and the explanation likely does not reflect the true reasoning process. One major strength of this approach is that it avoids several pitfalls associated with context perturbation methods, such as the model reconstructing the modified context. However, the authors themselves note that high recall cannot

be guaranteed: if unlearning a reasoning step does not alter the prediction, this may indicate failed unlearning, the existence of multiple reasoning paths, or the possibility that the step was not actually used. Furthermore, the FUR approach is not applicable to closed-source API-only models.

2.3 Causal Concept Faithfulness Metric

2.3.1 Mathematical Setting

Problem Setting

The goal is to evaluate the faithfulness of explanations generated by an opaque large language model (LLM) under evaluation, denoted by M , for a dataset of questions $X = \{x_1, \dots, x_N\}$. Matton et al. formulate their metric for so-called *context-based questions*, where each response r produced by the model consists of both a selected answer $y \in Y$ from a predefined set of answer options and a natural language explanation e , i.e.,

$$r = (y, e).$$

The underlying assumption is that the explanation implicitly or explicitly indicates which parts of the input influenced the model’s prediction. Rather than considering these parts as individual tokens or words, Matton et al. model them as high-level *concepts*. Consequently, the proposed metric measures the discrepancy between “the set of concepts that LLM explanations imply are influential and the set that truly are” [24].

Concepts and Categories

For each question x , a set of concepts

$$C = \{C_1, \dots, C_M\}$$

is identified. Unlike token-based approaches, concepts correspond to semantically meaningful properties of the input rather than low-level textual features. Each concept C_m can assume different possible values $c_m \in \mathcal{C}_m$, where $\mathcal{C}_m$ denotes the domain of all plausible values for concept C_m .

A key assumption of the metric—which will later be challenged in this thesis—is that concepts are “disentangled”. In other words, each concept can be modified independently without affecting the remaining concepts. This assumption allows causal interventions to be applied to individual concepts while keeping all others fixed.

Furthermore, every concept is assigned to a higher-level category from the set

$$K = \{K_1, \dots, K_L\},$$

for example *gender*, *ethnicity*, or *medical symptoms*. Besides improving the interpretability of the resulting analysis, these categories are also required for the Bayesian

hierarchical model employed by Matton et al. to estimate causal concept effects through partial pooling.

Causal Concept Effects (CE)

To estimate the actual influence of a concept on the model’s prediction, Matton et al. generate counterfactual questions. A counterfactual,

$$x_{c_m \rightarrow c'_m},$$

is obtained by replacing the original value c_m of concept C_m with an alternative value c'_m , while leaving the remainder of the input unchanged. Let

$$\mathcal{C}'_m = \mathcal{C}_m \setminus \{c_m\}$$

denote the set of all alternative values of concept C_m .

The causal concept effect is then defined as the average Kullback–Leibler divergence between the model’s prediction distribution for the original question and each corresponding counterfactual:

$$CE(x, C_m) = \frac{1}{|\mathcal{C}'_m|} \sum_{c'_m \in \mathcal{C}'_m} D_{\text{KL}} \left(P_M(Y \mid x_{c_m \rightarrow c'_m}) \parallel P_M(Y \mid x) \right). \quad (2.1)$$

Intuitively, concepts whose modification substantially changes the model’s output receive a high causal concept effect, whereas concepts with little influence receive values close to zero.

Explanation-Implied Effects (EE)

While the causal concept effect measures the actual influence of a concept on the model’s prediction, the explanation-implied effect estimates how influential the model itself claims the concept to be within its explanation. If the explanations are faithful, concepts with large causal effects should be mentioned more frequently than concepts with only a negligible influence.

Matton et al. therefore define the explanation-implied effect as the probability that the model’s explanations for both the original question and its counterfactuals identify concept C_m as being causally relevant [24]:

$$EE(x, C_m) = \frac{1}{|\mathcal{C}_m|} \sum_{c'_m \in \mathcal{C}_m} P_M(C_m \in E \mid x_{c_m \rightarrow c'_m}). \quad (2.2)$$

In practice, this quantity is estimated by automatically extracting the concepts referenced in the generated explanations using an auxiliary LLM.

2.3.2 Causal Concept Faithfulness

After computing both the causal concept effects and the explanation-implied effects for every concept of a question, the final faithfulness score can be determined. Specifically, the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects is calculated:

$$F(x) = \text{PCC}(CE(x, C), EE(x, C)). \quad (2.3)$$

A high positive correlation indicates that concepts which genuinely influence the model’s prediction are also consistently identified as influential within the explanation, whereas a low or even negative correlation indicates unfaithfulness.

Finally, the dataset-level faithfulness of model M is obtained by averaging the question-level faithfulness scores over all questions in the dataset:

$$F(X) = \frac{1}{|X|} \sum_{x \in X} F(x). \quad (2.4)$$

2.3.3 Evaluation Pipeline

Causal Concept Definition

As discussed before, Matton et al. metric of causal concept faithfulness is based on the idea of measuring the difference between “the set of concepts that LLM explanations imply are influential and the set that truly are” [24]. For understanding how this metric works in detail, the evaluation pipeline is presented in the following step by step while keeping the notation introduced by Matton et al.

There is a model M which is an opaque black-box and a set of questions $X = \{x_1, \dots, x_N\}$. We can only then access discrete samples of the probability distribution $\mathbb{P}_M(R | x)$ of M to a question x .

2.3.4 The Case for this Metric

2.4 The Correlated Concepts Problem

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